

AquaPress AP-600 Differential Pressure Transmitter -- Technical Manual

PRODUCT OVERVIEW

The AquaPress AP-600 is a state-of-the-art differential pressure transmitter designed specifically for filter backwash triggering in water treatment plants. This robust and reliable device provides accurate differential pressure measurements, allowing facility operators to maintain optimal filtration performance and ensure consistent water quality. With its advanced sensor technology, the AP-600 offers real-time monitoring, ensuring timely backwashing of filters to prevent clogging and maintain system efficiency. The device features a durable housing, easy installation, and compatibility with standard control systems, making it an ideal choice for modern water treatment facilities.

TECHNICAL SPECIFICATIONS

- Model: AquaPress AP-600
- Measurement Range: 0 to 100 psi (0 to 689.5 kPa)
- Accuracy: $\pm 0.25\%$ of full scale
- Output Signal: 4-20 mA, HART protocol compatible
- Power Supply: 24 VDC $\pm 10\%$
- Ambient Temperature Range: -40°F to 185°F (-40°C to 85°C)
- Wetted Materials: Stainless Steel 316L, Viton O-rings
- Housing Material: Polycarbonate, IP67 rated
- Dimensions: 6.5 in x 4.0 in x 3.5 in (165 mm x 102 mm x 89 mm)
- Weight: 2.5 lbs (1.13 kg)
- Mounting: 2-inch NPT or flange mount options available

CALIBRATION PROCEDURE

To ensure optimal performance of the AquaPress AP-600, regular calibration is essential. Calibration involves two primary adjustments: zero and span adjustments for differential pressure.

Zero Adjustment:

1. Disconnect the AP-600 from any pressure source.
2. Power on the device and allow it to stabilize for at least 5 minutes.
3. Access the transmitter's configuration settings using a HART communicator or the local display interface.
4. Navigate to the zero adjustment menu.
5. Set the zero point to 0 psi (0 kPa) and confirm the adjustment.
6. Verify the zero setting by applying a known zero pressure (using a vacuum source) and ensure the output reads 4 mA.

Span Adjustment:

1. Connect a calibrated pressure source to the high-pressure port of the AP-600.

2. Apply a known reference pressure, ideally at the midpoint of the measurement range (e.g., 50 psi).
3. Access the span adjustment menu using your HART communicator or local display.
4. Adjust the span setting to ensure the output reads 12 mA at 50 psi.
5. Repeat the process for the full-scale pressure of 100 psi, confirming that the output reads 20 mA.
6. Save all changes and exit the configuration settings.

ERROR CODES

The AquaPress AP-600 is equipped with a built-in diagnostic system that alerts users of potential issues via error codes. Below are common error codes, their causes, and recommended resolutions:

- Error Code: E1 - SENSOR FAULT

Cause: Internal sensor malfunction or failure.

Resolution: Inspect the wiring and connections; replace the sensor if necessary.

- Error Code: E2 - ZERO OFFSET ERROR

Cause: The zero point is out of specification.

Resolution: Perform a zero adjustment calibration procedure as described above.

- Error Code: E3 - SPAN ERROR

Cause: The span calibration is incorrect.

Resolution: Execute the span adjustment calibration procedure to correct the output.

- Error Code: E4 - OVER-RANGE

Cause: Input pressure exceeds the maximum measurement range.

Resolution: Reduce the pressure to within the specified range; check the system for blockages.

- Error Code: E5 - UNDER-RANGE

Cause: Input pressure is below the minimum measurement range.

Resolution: Ensure that the system has sufficient pressure; verify connections for leaks.

MAINTENANCE

Regular maintenance of the AquaPress AP-600 is crucial to ensure long-term reliability and performance. Follow these guidelines to maintain your device:

1. **Visual Inspection**: Conduct monthly visual inspections of the transmitter for signs of wear, corrosion, or damage. Check all connections for tightness and leaks.
2. **Cleaning**: Clean the exterior of the device with a soft cloth and mild detergent. Avoid using abrasive materials that may scratch the housing.
3. **Calibration**: Perform calibration every six months or as required by your facility's quality assurance protocols to ensure accurate readings.
4. **Firmware Updates**: Regularly check for firmware updates from AquaTech Instruments and apply them as necessary to improve performance and reliability.
5. **Record Keeping**: Maintain a log of all maintenance activities, including calibration dates, any repairs

performed, and parts replaced to ensure compliance with operational standards.

By adhering to these procedures, the AquaPress AP-600 will provide reliable service and help maintain optimal water treatment processes at your facility.